

Market Data
(options chain)



SVI Smoothing
(per-expiration IV fit)



Log-Moneyness Transform
($k = \log(K/F)$)



GP Derivative Estimation
(analytical kernel derivatives)



SINDy Sparse Regression
(identify active PDE terms)



KAN Nonlinear Diagnostics
(learn operator on active terms)



Validation
(PINN + OoS + misspecification)